Reliable Data Transfer Protocol over UDP

Output Screenshots:

1.python3 test_transfer.py

```
saran@Sarans-MacBook-Air rdt_udp % python3 test_transfer.py
SAW w=1  {}                                                                    match=True  time=0.00s retx=0 dup=0 ooo=0 corrup
t=0
SAW w=1  {'loss': 0.15, 'corrupt': 0.05, 'delay': 0.005, 'jitter': 0.01}       match=True  time=3.01s retx=31 dup=13 ooo=0 corr
upt=6
GBN w=8  {}                                                                    match=True  time=0.01s retx=0 dup=0 ooo=0 corrup
t=0
GBN w=8  {'loss': 0.15, 'corrupt': 0.05, 'delay': 0.005, 'jitter': 0.01}       match=True  time=2.90s retx=316 dup=2 ooo=240 co
rrupt=18
SR  w=8  {}                                                                    match=True  time=0.01s retx=0 dup=0 ooo=0 corrup
t=0
SR  w=8  {'loss': 0.15, 'corrupt': 0.05, 'delay': 0.005, 'jitter': 0.01}       match=True  time=0.82s retx=31 dup=13 ooo=34 cor
rupt=6

ALL TESTS PASSED
```

2.python3 receiver.py --port 9000 --out received.bin

```
=== Receiver summary ===
Saved to         : received.bin (150000 bytes, 147 packets)
Delivered in order : 147
Duplicates dropped : 0
Out-of-order       : 54  (buffered: 54)
Corrupted dropped  : 0
SHA-256            : 1657a7f7e203d4c421dc5954c2ad3ca7e91942eeb51ee72153bcdd5e60120f60
```

3. python3 sender.py sample.bin --port 9000 --mode sr --window 8 --timeout 0.05 --loss 0.1

```
=== Sender summary ===
Mode               : SR (window=8)
File               : 150000 bytes in 147 packets
Data packets sent  : 161
Retransmissions    : 14  (timeouts: 14)
ACKs received      : 147  (corrupt ACKs: 0)
Simulated network  : 14 dropped, 0 corrupted of 163 sent
Transfer time      : 0.614 s
Throughput         : 238.6 KB/s
SHA-256            : 1657a7f7e203d4c421dc5954c2ad3ca7e91942eeb51ee72153bcdd5e60120f60
saran@Sarans-MacBook-Air rdt_udp %
 * History restored
```

4. shasum -a 256 sample.bin received.bin

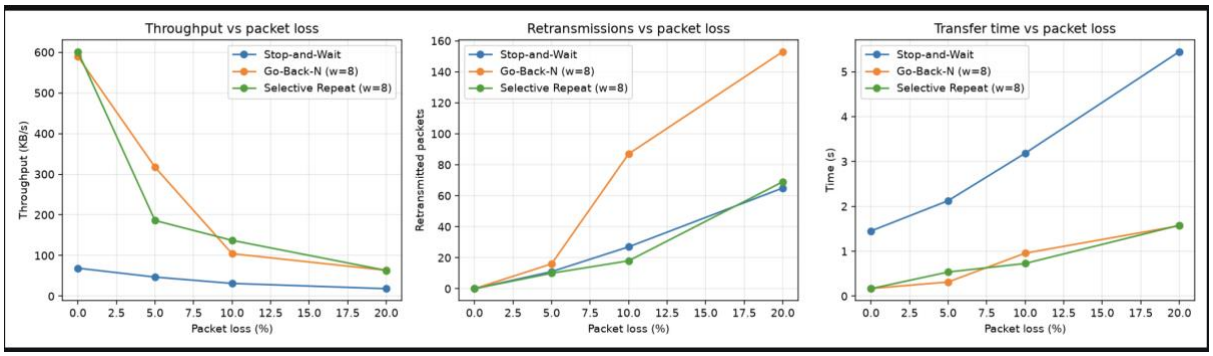
```
saran@Sarans-MacBook-Air rdt_udp % shasum -a 256 sample.bin received.bin
1657a7f7e203d4c421dc5954c2ad3ca7e91942eeb51ee72153bcdd5e60120f60  sample.bin
1657a7f7e203d4c421dc5954c2ad3ca7e91942eeb51ee72153bcdd5e60120f60  received.bin
```

5. python3 benchmark.py

```
saran@Sarans-MacBook-Air rdt_udp % python3 benchmark.py
{'experiment': 1, 'mode': 'saw', 'window': 1, 'loss_pct': 0, 'time_s': 1.452, 'throughput_kBps': 68.9, 'retransmissions': 0.0}
{'experiment': 1, 'mode': 'gbn', 'window': 8, 'loss_pct': 0, 'time_s': 0.17, 'throughput_kBps': 589.2, 'retransmissions': 0.0}
{'experiment': 1, 'mode': 'sr', 'window': 8, 'loss_pct': 0, 'time_s': 0.166, 'throughput_kBps': 601.4, 'retransmissions': 0.0}
{'experiment': 1, 'mode': 'saw', 'window': 1, 'loss_pct': 5, 'time_s': 2.126, 'throughput_kBps': 47.0, 'retransmissions': 11.0}
{'experiment': 1, 'mode': 'gbn', 'window': 8, 'loss_pct': 5, 'time_s': 0.315, 'throughput_kBps': 317.3, 'retransmissions': 16.0}
{'experiment': 1, 'mode': 'sr', 'window': 8, 'loss_pct': 5, 'time_s': 0.537, 'throughput_kBps': 186.2, 'retransmissions': 10.0}
{'experiment': 1, 'mode': 'saw', 'window': 1, 'loss_pct': 10, 'time_s': 3.184, 'throughput_kBps': 31.4, 'retransmissions': 27.0}
{'experiment': 1, 'mode': 'gbn', 'window': 8, 'loss_pct': 10, 'time_s': 0.957, 'throughput_kBps': 104.5, 'retransmissions': 87.0}
{'experiment': 1, 'mode': 'sr', 'window': 8, 'loss_pct': 10, 'time_s': 0.728, 'throughput_kBps': 137.3, 'retransmissions': 18.0}
{'experiment': 1, 'mode': 'saw', 'window': 1, 'loss_pct': 20, 'time_s': 5.448, 'throughput_kBps': 18.4, 'retransmissions': 65.0}
{'experiment': 1, 'mode': 'gbn', 'window': 8, 'loss_pct': 20, 'time_s': 1.569, 'throughput_kBps': 63.7, 'retransmissions': 153.0}
{'experiment': 1, 'mode': 'sr', 'window': 8, 'loss_pct': 20, 'time_s': 1.579, 'throughput_kBps': 63.3, 'retransmissions': 69.0}
{'experiment': 2, 'mode': 'gbn', 'window': 1, 'loss_pct': 5, 'time_s': 2.118, 'throughput_kBps': 47.2, 'retransmissions': 11.0}
{'experiment': 2, 'mode': 'sr', 'window': 1, 'loss_pct': 5, 'time_s': 2.121, 'throughput_kBps': 47.2, 'retransmissions': 11.0}
{'experiment': 2, 'mode': 'gbn', 'window': 2, 'loss_pct': 5, 'time_s': 1.15, 'throughput_kBps': 87.0, 'retransmissions': 12.0}
{'experiment': 2, 'mode': 'sr', 'window': 2, 'loss_pct': 5, 'time_s': 1.405, 'throughput_kBps': 71.2, 'retransmissions': 12.0}
{'experiment': 2, 'mode': 'gbn', 'window': 4, 'loss_pct': 5, 'time_s': 0.776, 'throughput_kBps': 128.8, 'retransmissions': 24.0}
{'experiment': 2, 'mode': 'sr', 'window': 4, 'loss_pct': 5, 'time_s': 0.905, 'throughput_kBps': 110.5, 'retransmissions': 12.0}
{'experiment': 2, 'mode': 'gbn', 'window': 8, 'loss_pct': 5, 'time_s': 0.316, 'throughput_kBps': 316.7, 'retransmissions': 16.0}
{'experiment': 2, 'mode': 'sr', 'window': 8, 'loss_pct': 5, 'time_s': 0.539, 'throughput_kBps': 185.4, 'retransmissions': 10.0}
{'experiment': 2, 'mode': 'gbn', 'window': 16, 'loss_pct': 5, 'time_s': 0.384, 'throughput_kBps': 260.7, 'retransmissions': 64.0}
{'experiment': 2, 'mode': 'sr', 'window': 16, 'loss_pct': 5, 'time_s': 0.343, 'throughput_kBps': 291.4, 'retransmissions': 11.0}
{'experiment': 2, 'mode': 'gbn', 'window': 32, 'loss_pct': 5, 'time_s': 0.275, 'throughput_kBps': 363.4, 'retransmissions': 93.0}
{'experiment': 2, 'mode': 'sr', 'window': 32, 'loss_pct': 5, 'time_s': 0.229, 'throughput_kBps': 436.3, 'retransmissions': 6.0}

Saved results.csv, graph_loss_comparison.png, graph_window_size.png
saran@Sarans-MacBook-Air rdt_udp %
```

Screenshot of graph loss comparision:



Screenshot of window size

